

Quartet Reduction and Sharp Degree Bounds at the Last Two Capacities

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Abstract

For $N \geq 3$ distinct scalar posterior states with positive masses, we study the hierarchy price at capacities $(N - 2, N - 1)$. For every twice continuously differentiable strictly convex generator, either compatible flat optima exist, or a restriction to four consecutive original states has the same two flat costs and a deterministic hierarchy price at least as large. The hierarchy class includes noncontiguous cells. Consequently, for each fixed generator, the deterministic envelope over all finite source counts equals the weighted quartet envelope. The inherited polynomial quartet estimate gives the uniform upper bound $1936r^2/(\log r)^2$ for curvature degree at most $r \geq 2$. An exterior-anchor construction embeds the inherited lower witnesses into every prescribed $N \geq 4$ and preserves their deterministic prices. Both deterministic and stochastic degree envelopes therefore have order $\Theta(r^2/(\log r)^2)$, with constants and a degree threshold independent of N . The witnesses admit interior posterior support and smooth proper losses bounded in $[0, 1/2]$, while their absolute flat excess risks vanish. Exact finite checks cover 300 instances and 67,860 compatible hierarchy candidates. The result concerns only the last two capacities; arbitrary prescribed capacities and simultaneous all-capacity hierarchies remain outside its scope.

1 Question, contribution, and quantifiers

An optimal representation with k outputs need not be obtainable by coarsening an optimal representation with $k + 1$ outputs. For a scalar posterior and a strictly proper binary loss, the excess Bayes risk of a representation is a Jensen distortion. The hierarchy price measures the smallest simultaneous multiplicative cost of requiring this coarsening. We study the two capacities immediately below exact representation.

The preceding weighted four-state analysis [3] proved the sharp degree order at capacities $(2, 3)$, with no lower bounds on masses or posterior gaps. The additional argument here is combinatorial: at capacities $(N - 2, N - 1)$, the possible flat partitions leave only a consecutive quartet as an obstruction to compatibility. Singleton extension preserves both flat denominators and costs. For the lower bound, expensive exterior anchors exclude every new merge, and an affine change restores legal posterior coordinates without changing the price.

Let $C_{N,r}^{\det}$ and $C_{N,r}^{\text{stoch}}$ be the suprema of the respective prices over N distinct locations in $[0, 1]$, positive probabilities, and generators whose nonnegative nonzero polynomial curvature has degree at most r . The generator, weights, and locations may all vary with N, r . The results below give an absolute r_0 such that, for every $N \geq 4$ and every integer $r \geq r_0$,

$$\frac{1}{9216} \frac{r^2}{(\log r)^2} \leq C_{N,r}^{\text{stoch}} \leq C_{N,r}^{\det} \leq 1936 \frac{r^2}{(\log r)^2}. \quad (1)$$

For $N = 3$, both prices are one. All logarithms are natural; r is the degree of the curvature, not of the generator.

Three quantifiers must be distinguished. The fixed-generator identity is an equality between the supremum over all finite N and the quartet supremum. A general individual source is only dominated by a selected quartet, and the inequality can be strict. The fixed- N lower witnesses are a separate construction in the polynomial degree class; they do not give fixed-generator envelope equality for each N , or exact stochastic price preservation.

2 Model and elementary facts

Fix $N \geq 3$, locations $x_1 < \dots < x_N$ in an interval I , and weights $p_i > 0$. The weights may sum to any positive number. Let φ be C^2 and strictly convex on I , and write $g = \varphi'' \geq 0$. For a nonempty cell $C \subseteq \{1, \dots, N\}$, set

$$p_C = \sum_{i \in C} p_i, \quad m_C = p_C^{-1} \sum_{i \in C} p_i x_i, \quad D(C) = \sum_{i \in C} p_i \varphi(x_i) - p_C \varphi(m_C).$$

For a partition P , set $D(P) = \sum_{C \in P} D(C)$, and let OPT_k be the minimum of $D(P)$ over partitions with at most k cells. The deterministic price at the capacities in question is

$$R_N = \min_{H_{N-1} \succeq H_{N-2}} \max \left\{ \frac{D(H_{N-1})}{\text{OPT}_{N-1}}, \frac{D(H_{N-2})}{\text{OPT}_{N-2}} \right\}, \quad |H_j| \leq j.$$

Write R_N^{stoch} for the stochastic analogue defined below. The notation $\rho_{N,\text{det}} = R_N$ and $\rho_{N,\text{stoch}} = R_N^{\text{stoch}}$ is used interchangeably in constructions; subscripts may instead indicate the generator, and superscripts may display the capacity pair. An omitted source count in the inherited endpoint construction means four states with capacities $(2, 3)$. Refinement is denoted by $\succeq$. No contiguity restriction is imposed on either hierarchy partition. Strict convexity, positive weights, and distinct locations imply $\text{OPT}_k > 0$ for $k < N$, so all displayed ratios are well defined. Multiplying every weight by one common positive factor multiplies every distortion by that factor and leaves every price unchanged.

Lemma 2.1 (Contiguous flat optima and deficit forms). *A contiguous flat optimum exists for every $k \leq N$, and every flat optimum fills its budget when $k < N$. In particular, a contiguous optimum at capacity $N - 1$ consists of one adjacent pair and all remaining singletons. A contiguous optimum at capacity $N - 2$ consists of either one consecutive triple or two disjoint adjacent pairs, with all remaining states singletons. Therefore*

$$q := \text{OPT}_{N-1} = \min_{1 \leq i < N} D(\{i, i+1\}).$$

Proof. The centroid identity for the Bregman divergence $B_\varphi(x, a) = \varphi(x) - \varphi(a) - \varphi'(a)(x - a)$ is

$$\sum_{i \in C} p_i B_\varphi(x_i, a) = D(C) + p_C B_\varphi(m_C, a).$$

Thus a weighted centroid minimizes the reporting cost of its cell. For ordered distinct centers $a < b$, the difference $B_\varphi(x, a) - B_\varphi(x, b)$ is affine in x with positive slope $\varphi'(b) - \varphi'(a)$. Assigning each source point to a nearest center, with ordered tie-breaking, consequently gives contiguous cells. Starting with centers of a flat optimum, this reassignment and subsequent centroid updates cannot increase its distortion, and use at most k cells. Duplicate centers, if any, may first be identified. This proves existence of a contiguous optimum.

Splitting a nonsingleton cell into its smallest-location singleton and the other states strictly decreases its cost: their means are distinct, so the corresponding two-group Jensen gap is strictly positive. Any partition with fewer than $k < N$ cells can therefore be improved. Finally, the deficit identity $N - |P| = \sum_{C \in P} (|C| - 1)$ gives the two asserted forms. Every adjacent pair supplies a feasible $(N - 1)$ -cell partition, which proves the formula for q . $\square$

Lemma 2.2 (Two adjacent costs inside a triple). *For three ordered states $A < B < C$, with their original positive weights,*

$$D(ABC) \geq D(AB) + D(BC).$$

Proof. For any cell, the Jensen kernel is

$$K_C(t) = \sum_{i \in C} p_i(x_i - t)_+ - p_C(m_C - t)_+, \quad D(C) = \int_I K_C(t)g(t) dt.$$

The kernel identity follows by integrating φ'' twice; affine terms cancel. For each fixed t , the hinge $x \mapsto (x - t)_+$ is convex. The Jensen gap of a union equals the sum of the within-group Jensen gaps plus the nonnegative Jensen gap of the group centroids. Applying this to AB and C gives $K_{ABC} \geq K_{AB}$; applying it to A and BC gives $K_{ABC} \geq K_{BC}$. The two pair kernels have disjoint open supports, (A, B) and (B, C) , and vanish at the common endpoint. Hence pointwise $K_{ABC} \geq K_{AB} + K_{BC}$. Integrate against $g \geq 0$. $\square$

2.1 Stochastic hierarchies and exact cardinalities

For probability weights, a finite channel $X \rightarrow Z$ has distortion

$$D(Z) = \mathbb{E}\varphi(x_X) - \mathbb{E}\varphi(\mathbb{E}[x_X | Z]).$$

A stochastic hierarchy is a degradation chain $X \rightarrow Z_{N-1} \rightarrow Z_{N-2}$, with each output alphabet of size at most its subscript. Its price R_N^{stoch} uses the same two flat denominators. Indeed, sample an independent deterministic encoder table U realizing any channel. Revealing the table and applying conditional Jensen gives $D(Z) \geq \mathbb{E}_U D(Z | U) \geq \text{OPT}_k$. A deterministic flat minimizer attains the reverse inequality. Thus allowing stochastic flat encoders does not lower OPT_k . General positive weights are handled by common normalization.

Lemma 2.3 (Inherited two-level deterministic comparison). *For this finite-source problem,*

$$\frac{1}{2}R_N \leq R_N^{\text{stoch}} \leq R_N.$$

Proof. Given a stochastic degradation chain, independently sample deterministic function tables for its encoder and coarsener, independently of X , and denote the two tables together by U . Conditional on U they define an admissible deterministic hierarchy. Revealing U decreases Jensen distortion at each level, so $D(Z_k) \geq \mathbb{E}_U D(Z_k | U)$. If the stochastic chain has price R , the expected sum of the two conditional deterministic ratios is at most $2R$. Some table has sum, and hence maximum ratio, at most $2R$. Taking the infimum over chains gives $R_N \leq 2R_N^{\text{stoch}}$. Deterministic hierarchies are admissible stochastic chains, proving the other inequality. The table is a proof device, not a public random seed available to the predictor. This is the Phase 9 comparison [3]. $\square$

Lemma 2.4 (Filling both hierarchy budgets). *Any hierarchy with at most $N - 1$ fine cells and at most $N - 2$ coarse cells can be refined to those exact cardinalities without increasing either distortion.*

Proof. Split fine cells within their current coarse cells until the fine level has $N - 1$ cells. If the coarse level has fewer than $N - 2$ cells, at least one coarse cell contains two fine cells; split that coarse cell into unions of fine cells. Repeat until its budget is filled. Jensen decomposition makes each refinement nonincreasing in cost, and nesting is maintained. No contiguous-cell condition is used. $\square$

The strict-convexity assumptions make the relevant flat optima positive. Repeated locations, zero weights, and identically zero curvature are not included in the theorems; such extensions would require a convention for zero-optimum ratios.

3 The exact restriction theorem

Theorem 3.1 (Consecutive quartet domination). *Choose any cheapest adjacent pair e of cost $q = \text{OPT}_{N-1}$ and any contiguous flat optimum P at capacity $N - 2$, of cost $F = \text{OPT}_{N-2}$. One of the following conclusions holds.*

- (i) *There is a hierarchy with both levels flat optimal, so $R_N = 1$.*
- (ii) *The only nonsingleton cells of P are two pairs AB and CD on four consecutive original states $A < B < C < D$, and $e = BC$. For the source restricted to $S = \{A, B, C, D\}$, keeping its original weights, its flat values satisfy*

$$\text{OPT}_3(S) = q, \quad \text{OPT}_2(S) = F = D(AB) + D(CD),$$

and its unrestricted deterministic $(2, 3)$ hierarchy price satisfies

$$R_N \leq R_4(S).$$

In particular, if $R_N > 1$, conclusion (ii) holds for every such choice of e and P . When $N = 3$, necessarily $R_N = 1$.

Proof. The partition whose sole nonsingleton cell is e is flat optimal at the fine capacity. We first check all possible forms of P .

Suppose the sole nonsingleton cell of P is a triple $T = ABC$ of consecutive states. If both endpoints of e belong to T , the fine partition just described refines P , proving (i). If e is disjoint from T , take $f = AB$ and $h = BC$. Then e and f are disjoint, and

$$D(e) + D(f) \leq D(h) + D(f) \leq D(T) = F,$$

because e is globally cheapest among adjacent pairs and Lemma 2.2 applies. The partition with nonsingleton cells e, f has $N - 2$ cells and cost at most F , hence exactly F . It is a flat optimum compatible with the chosen fine partition. If e meets T in exactly one state, adjacency of e and consecutivity of T force that intersection to be an endpoint of T . If the endpoint is A , take $f = BC$ and $h = AB$; if it is C , take $f = AB$ and $h = BC$. Again e is disjoint from f , and the same inequality proves (i). These cases exhaust the triple form.

Suppose instead that the only nonsingleton cells of P are disjoint adjacent pairs f and h , with $F = D(f) + D(h)$. If e equals either of them, the chosen fine partition refines P . If e is disjoint from at least one of them, say f , then the partition with nonsingleton cells e, f is feasible at capacity $N - 2$, and

$$D(e) + D(f) \leq D(h) + D(f) = F.$$

It is therefore a compatible flat optimum. Notice that this argument allows e to meet h , or to be disjoint from both f and h .

The sole case remaining is that e meets both f and h . Write the two coarse pairs in order as AB and CD . An adjacent pair meeting both must be BC . Thus A, B, C, D are four consecutive original states, and P and e have the form in (ii).

It remains to prove the exact flat-value identities and the price comparison. The restricted quartet has a three-cell partition with pair BC , of cost q , and the two-cell partition $AB \mid CD$, of cost F . Conversely, any partition of S with at most three (respectively two) cells extends to a partition of the entire source with at most $N - 1$ (respectively $N - 2$) cells by making every state outside S a singleton. Its distortion is unchanged because every outside singleton has zero cost. This rules out restricted values smaller than q or F and proves the two identities. This step applies to arbitrary restricted partitions, including noncontiguous ones.

Finally, apply the same singleton extension simultaneously to both levels of any restricted hierarchy. Refinement is preserved, distortion is unchanged, and the denominators are precisely the same. The full-source optimum can only be better than this feasible hierarchy, proving $R_N \leq R_4(S)$. Restricted optimal hierarchies exist because their partition set is finite; equivalently one can take the infimum of the same comparison over all of them. For $N = 3$ the coarse capacity is one, whose single cell is compatible with every fine partition. $\square$

Corollary 3.2 (Equality of the all-size and quartet envelopes). *For a fixed C^2 strictly convex generator φ on I , let $\mathcal{C}_N(\varphi)$ be the supremum of R_N over N distinct locations in I and positive weights. Then*

$$\mathcal{C}_N(\varphi) \leq \mathcal{C}_4(\varphi) \quad (N \geq 3), \quad \sup_{N \geq 3} \mathcal{C}_N(\varphi) = \mathcal{C}_4(\varphi).$$

The same equality holds after taking a supremum over any class of such generators, in particular nonnegative nonzero polynomial curvatures of degree at most r on $[0, 1]$.

Proof. Every hierarchy price is at least one, and the quartet envelope is at least one. Apply Theorem 3.1 pointwise. Taking the supremum over N includes $N = 4$, so the reverse inequality for the last display is automatic. If probabilities summing to one are required, divide every weight in the restricted quartet by the same total mass; the price is unchanged by this common scaling. No individual cell is conditionally renormalized. $\square$

Remark 3.3 (Meaning and limits of “exact”). The reduction has no multiplicative loss and preserves both flat denominators exactly before any common weight normalization. It asserts domination by a restriction, not equality of an individual source price with the selected quartet price. The lifted quartet hierarchies form only a subset of the global feasible hierarchies. The corollary asserts equality of the supremum over all finite source counts with the quartet supremum; it does not prove equality separately for every fixed $N > 4$ and fixed generator. Section 5 instead constructs special polynomial witnesses with exact deterministic price preservation.

4 The uniform polynomial upper bound

Theorem 4.1 (Uniform degree bound at the top two capacities). *Let $N \geq 3$, let $x_1 < \dots < x_N$ lie in $[0, 1]$ with arbitrary positive weights, and suppose $g = \varphi''$ is a nonzero polynomial of degree at most r , nonnegative on $[0, 1]$. For the deterministic price R_N at capacities $(N - 2, N - 1)$:*

$$\begin{aligned} R_N \geq 128, \quad r \geq 1 &\implies R_N^2 \leq 16r^2 \exp\left(\frac{20r}{\sqrt{R_N}}\right), \\ R_N \geq 256, \quad r \geq 1 &\implies R_N(\log R_N)^2 \leq 1936r^2. \end{aligned}$$

For every integer $r \geq 2$,

$$R_N \leq 1936 \frac{r^2}{(\log r)^2}.$$

The constant is independent of the source count, all positive weights, and all positive posterior gaps. The inherited Phase 9 matching quartet lower construction makes the supremum over all $N \geq 3$ have sharp order $\Theta(r^2/(\log r)^2)$.

Proof. A nonzero nonnegative polynomial vanishes on no interval, so its twice-antiderivative is strictly convex. Apply Theorem 3.1. There is nothing to prove if $R_N = 1$; otherwise $R_N \leq R_4(S)$ for the specified quartet. The weighted quartet bound, proved in Phase 9 and reproduced in Proposition A.1, gives the final displayed estimate. For the implicit inequality, the quartet estimate can be written

$$R_4(S)^2 \exp\left(-\frac{20r}{\sqrt{R_4(S)}}\right) \leq 16r^2.$$

The function $s \mapsto s^2 \exp(-20r/\sqrt{s})$ is strictly increasing for $s > 0$, so this inequality transfers to R_N whenever $R_N \geq 128$. Likewise $s \mapsto s(\log s)^2$ is increasing for $s \geq 1$, which gives the second implication. The assertion about the envelope's lower order uses the stated, inherited quartet witnesses and the fact that the all-size supremum includes $N = 4$; no padding argument is used. $\square$

Remark 4.2 (Stochastic consequence of the restriction proof). Every deterministic hierarchy is an admissible stochastic hierarchy, with the same flat denominators. Thus the same uniform upper bound holds for the stochastic hierarchy price. The restriction proof itself is a deterministic argument: it does not establish an equality between the all-size stochastic envelope and the stochastic quartet envelope. The inherited stochastic quartet lower witnesses still establish the matching all-size stochastic degree order.

5 Sharp lower bounds at every source count

The restriction theorem is an upper argument. To obtain a lower witness at an arbitrarily prescribed source count, extra states must not create cheap merges. The globally positive curvature floor of the inherited endpoint construction permits a direct exterior-anchor embedding. Its auxiliary coordinates may be arbitrarily far apart; an affine change places every posterior in the required interior interval afterward.

Theorem 5.1 (Uniform-in- N matching lower witnesses). *There are absolute constants $c > 0$ and r_0 such that, for every integer $N \geq 4$ and every integer $r \geq r_0$, there is a source with exactly N distinct posteriors in $[1/4, 3/4]$, all source weights positive, and a polynomial generator with positive curvature of degree at most r , whose induced smooth strictly proper losses are bounded in $[0, 1/2]$ on the full report interval $[0, 1]$, for which*

$$\rho_{\det}^{(N-2, N-1)} \geq \rho_{\text{stoch}}^{(N-2, N-1)} \geq c \frac{r^2}{(\log r)^2}.$$

For example, $c = 1/9216$ suffices with a suitably large absolute r_0 . The masses, generator, and posterior locations may depend on both N and r ; no minimum posterior gap is asserted.

The construction also preserves the original quartet deterministic price exactly; no exact stochastic price equality is asserted.

Proof. Construction and anchor penalty. Use the endpoint quartet of Appendix B, writing

$$d = (4 \log n/n)^2, \quad b = d^2, \quad a = 1/2 - b, \quad (x_A, x_B, x_C, x_D) = (0, 2d, 1 - 2d, 1),$$

and $g_n = \varphi_n'' = \kappa_n(t) + \kappa_n(1 - t) + b$. In particular $g_n(t) \geq b > 0$ for every real t , and its degree is exactly $2n$. Let $N \geq 5$, $m = N - 4$, and $\lambda = 1/2$. Give the original four states masses $\lambda(a, b, b, a)$ and give m additional anchors equal masses $\eta = 1/(2m)$. For all sufficiently large n , the smallest of these masses is

$$\mu = \min\{\lambda b, \eta\} > 0.$$

Set

$$L = \frac{2}{\sqrt{b\mu}}, \quad z_j = -jL \quad (1 \leq j \leq m).$$

This is an auxiliary real coordinate; the legal posterior coordinates are defined below. Every pair involving an anchor has gap at least L . The global curvature floor yields the strong-convexity Jensen estimate

$$\begin{aligned} D(uv) &\geq \frac{b}{2} \frac{p_u p_v}{p_u + p_v} (u - v)^2 \\ &\geq \frac{b \min(p_u, p_v)}{4} (u - v)^2 \geq \frac{b\mu L^2}{4} = 1. \end{aligned}$$

The first inequality follows by subtracting the convex quadratic $bx^2/2$ from the generator and applying Jensen to the remaining convex function. The other inequalities use $p_u p_v / (p_u + p_v) \geq \min(p_u, p_v)/2$. Thus the anchor penalty holds at every sufficiently large n admitted by the original quartet, with no additional N -dependent threshold.

Top-capacity obstruction. Let P, q, T, J denote the original quartet costs. Equations (15)–(17) give

$$bd/2 \leq P \leq 4bd, \quad 0 < q \leq bd^2, \quad T \geq b/4, \quad J \geq b/2,$$

and every core pair other than BC costs at least P . The augmented flat optima satisfy

$$0 < \text{OPT}_{N-1} \leq \lambda q, \quad 0 < \text{OPT}_{N-2} \leq 2\lambda P \leq 8\lambda bd,$$

by keeping all anchors singleton and using the quartet's fine or coarse candidate. Refine any deterministic hierarchy to use exactly $N - 1$ fine cells and $N - 2$ coarse cells; refinement cannot increase costs. The fine partition merges one pair. If it is not BC , its cost is at least λP : this holds for every other core pair, and an anchor pair costs at least one, exceeding λP for sufficiently large n . Its fine ratio is then at least $P/q \geq 1/(2d)$.

If the fine pair is BC , the coarse partition merges two fine blocks. A merge among core blocks has cost at least $\lambda b/4$ by the three original quartet coarsenings. Any merge involving an anchor contains a pair of cost at least one, hence also has cost at least one by the Jensen decomposition $D(S) \geq D(U)$ for $U \subset S$. Therefore its coarse ratio is at least

$$\frac{\lambda b/4}{8\lambda bd} = \frac{1}{32d}.$$

This proves the deterministic lower bound $1/(32d)$ for all $N \geq 5$; the $N = 4$ statement is the original quartet. Lemma 2.3 gives the stochastic lower bound $1/(64d)$ for every N . No assertion that optimal cells are contiguous is needed.

In fact, the deterministic quartet price is preserved exactly. Let q_4, Q_4 be the actual quartet flat optima at capacities three and two, and let C be the cost of merging the entire scaled core into one cell. For a cell of total mass λ supported on $[0, 1]$, its Jensen kernel satisfies

$$K(t) = \min \left\{ \sum_i p_i(t - x_i)_+, \sum_i p_i(x_i - t)_+ \right\} \leq \lambda \min(t, 1 - t) \leq \lambda/2.$$

Since $\int_0^1 g_n = 2 + b$ and $\lambda = 1/2$, we have $C \leq (2 + b)/4 < 1$ for all sufficiently large n . An augmented flat partition with a nonsingleton cell containing an anchor costs at least one, whereas a core-only candidate costs at most C . Thus

$$\text{OPT}_{N-1} = \lambda q_4, \quad \text{OPT}_{N-2} = \lambda Q_4.$$

Choose a core fine optimum and any compatible two-cell core coarsening. Its augmented hierarchy price is at most $C/(\lambda Q_4)$, because its fine ratio is one and $C \geq \lambda Q_4$. Any hierarchy whose coarse level mixes an anchor has ratio at least $1/(\lambda Q_4) > C/(\lambda Q_4)$; an anchor merged at the fine level is also merged at the coarse level. Therefore every optimal augmented deterministic hierarchy leaves every anchor singleton at both levels. Its remaining core hierarchy has capacities two and three, proving

$$\rho_{N,\text{det}} = \rho_{4,\text{det}}$$

for this embedding. This equality is special to the constructed sources; the upper restriction theorem for an arbitrary source asserts only domination. No corresponding exact stochastic equality is asserted.

Interior support and bounded loss. The auxiliary source hull is $[-mL, 1]$. Define

$$s = 2(1 + mL), \quad t = -mL - s/4, \quad x = sy + t, \quad M = \max_{x \in [t, t+s]} g_n(x).$$

Then $y = 1/4 + (x + mL)/[2(1 + mL)]$ maps the entire hull onto $[1/4, 3/4]$. The transformed generator

$$\tilde{\varphi}_n(y) = \frac{\varphi_n(sy + t)}{s^2 M}$$

has curvature $\tilde{g}(y) = g_n(sy + t)/M \in (0, 1]$ throughout $[0, 1]$, with unchanged degree $2n$. Every augmented-source distortion is multiplied by the same positive factor $1/(s^2 M)$, so all prices are preserved. Symmetry is not needed: define

$$\ell(q, 1) = \int_q^1 (1 - v) \tilde{g}(v) dv, \quad \ell(q, 0) = \int_0^q v \tilde{g}(v) dv.$$

Both losses lie in $[0, 1/2]$ on all of $[0, 1]$, are smooth, and the derivative of conditional risk is $(q - p)\tilde{g}(q)$ at posterior p . Therefore the loss is strictly proper and its excess risk is $B_{\tilde{\varphi}_n}(p, q)$.

Finally, put $n = \lfloor r/2 \rfloor$. For sufficiently large r , $n \geq r/3$ and $\log n \leq \log r$, so

$$\frac{1}{64d_n} = \frac{n^2}{1024(\log n)^2} \geq \frac{r^2}{9216(\log r)^2}.$$

All large-degree requirements above come from the original quartet or absolute inequalities such as $a \geq b$ and $\lambda P < 1$, proving that the same r_0 works for every finite N . There is no claim of a uniform positive gap or of nonvanishing absolute excess risk after normalization.

For explicit absolute-risk bookkeeping, the normalized flat costs obey

$$0 < \widetilde{\text{OPT}}_{N-1} \leq \frac{\lambda d^4}{s^2 M}, \quad 0 < \widetilde{\text{OPT}}_{N-2} \leq \frac{8\lambda d^3}{s^2 M}.$$

The interval $[t, t + s]$ contains $[0, 1]$, and $\int_0^1 g_n = 2 + b$, so $M \geq 2 + b$. Also $s \geq 2$ and $\lambda = 1/2$. Thus for $N \geq 5$ the respective bounds are at most $d^4/16$ and $d^3/2$, and both tend to zero uniformly in N . The raw anchor penalty of one is multiplied by the same $1/(s^2 M)$; it is not a positive absolute penalty after normalization. For $N = 4$, the normalization in Appendix B gives the respective bounds $d^4/8$ and d^3 . These weaker constants therefore give vanishing flat costs uniformly over all $N \geq 4$ as well. $\square$

6 Why tiny-mass padding is insufficient

Take $\varphi(x) = x^2$, initially with four locations

$$(A, B, C, D) = (0, 2, 3, 5)$$

and equal, unnormalized masses 1. Normalizing all masses multiplies every cost by the same factor and changes no price. Here

$$q = D(BC) = \frac{1}{2}, \quad Q = D(AB) + D(CD) = 4, \quad D(ABC) = D(BCD) = \frac{14}{3}.$$

The four-state deterministic price is exactly $7/6$. If the fine pair is BC , its two triple coarsenings have cost $14/3$, and its remaining coarsening $AD \mid BC$ has cost 13. Any other fine pair has cost at least 2, giving a fine-level ratio at least 4. This covers all exact hierarchies, including noncontiguous ones; hierarchies with unused capacity can be refined to exact capacities without increasing cost.

Now add $E = -1$ with mass $\varepsilon \in (0, 1)$, keeping the four original masses equal to 1, and normalize by $4 + \varepsilon$. At capacities three and four, both flat optima are attained by a single hierarchy:

$$H_4 = EA \mid B \mid C \mid D, \quad H_3 = EA \mid BC \mid D.$$

Its unnormalized costs are

$$q_5 = \frac{\varepsilon}{1 + \varepsilon} < \frac{1}{2}, \quad Q_5 = \frac{\varepsilon}{1 + \varepsilon} + \frac{1}{2} < 1.$$

The pair EA is strictly cheapest: other pairs touching E have the same mass coefficient and strictly larger squared distance, while all core pairs cost at least $1/2$.

It remains to check the coarse optimum. A three-cell partition of five states has one triple or two pairs. Every core triple costs at least $14/3$. A triple touching E and a core pair other than BC dominates that core pair and therefore costs at least 2. The remaining triple is EBC , of cost

$$D(EBC) = \frac{25\varepsilon + 1}{\varepsilon + 2}, \quad D(EBC) - Q_5 = \frac{\varepsilon(47\varepsilon + 45)}{2(\varepsilon + 1)(\varepsilon + 2)} > 0.$$

For two-pair partitions, if one pair is BC , the cheapest disjoint pair is EA . If neither pair is BC , at least one pair is a core pair of cost at least 2: two disjoint pairs cannot both touch E . These candidates also cost more than Q_5 . Thus the claimed coarse partition is optimal, and

$$\rho_{5,\det} = \rho_{5,\text{stoch}} = 1 \quad \text{for every } 0 < \varepsilon < 1.$$

All five locations can be moved into $[1/4, 3/4]$ by $y = (x + 4)/12$. Positive scaling makes the quadratic curvature at most one, and the usual curvature-integral construction gives strictly proper losses in $[0, 1/2]$. Thus neither distinctness, interior support, bounded loss, nor arbitrarily small padding mass rescues this failed embedding method.

This refutes preservation under naive tiny-mass padding. It does not contradict the carefully separated anchors in Theorem 5.1.

7 Exact finite checks and a strict restriction example

The finite checks use Python’s exact rational arithmetic, with no floating-point comparisons in the assertions. Their source and result files are under `research/experiments/`. They test the deterministic claim over a bounded grid; they do not prove a universal theorem, optimize stochastic chains, establish an asymptotic order numerically, or train a neural model.

The main check covers every combination of $N \in \{4, 5, 6, 7, 8\}$, three location grids, four weight patterns, and five curvature polynomials: 300 instances. The curvatures are $f^2 + 1/1000$, where

$$f \in \{1, 1 - 2t, 1 - 6t + 6t^2, 1 - 12t + 30t^2 - 20t^3, 1 - 20t + 90t^2 - 140t^3 + 70t^4\}.$$

Thus degrees are 0, 2, 4, 6, 8, and positivity is certified algebraically. The location grids are uniform, quadratic, and a fixed four-point core with additional ordered states. Weights are uniform, alternating 1 and 1/100, or have final four entries $(1, 10^{-4}, 10^{-4}, 1)$ or $(2, 10^{-2}, 10^{-4}, 1)$ with preceding entries one.

By Lemma 2.4, it suffices to cover exact-cardinality hierarchies. Each coarse partition has one triple or two disjoint pairs; the fine pair lies within one coarse cell. There are $\binom{N}{3} + 3\binom{N}{4}$ coarse shapes and $3\binom{N}{3} + 6\binom{N}{4}$ compatible coarse/fine candidates per instance, including every noncontiguous choice. The code minimizes the fine-pair cost within each coarse shape before taking the outer minimum, thereby covering all those compatible candidates without explicitly evaluating each maximum separately.

Recorded check	Count or outcome
Exact rational instances	300
Compatible hierarchy candidates covered	67,860
Cheapest-pair/coarse-optimum choices checked	807
Instances with deterministic price greater than one	49
Independent set-partition enumeration cross-check	$N = 4, 5, 6$
Maximum observed price	20000/10001
Tiny-mass padding values checked	5

For every optimal-choice combination, the assertions verify either a compatible optimum or a consecutive quartet, equality of its two flat costs, and domination of the global price. The independent restricted- growth-string enumerator checks the specialized coarse-shape enumeration for $N = 4, 5, 6$; the analytical deficit classification justifies it for the larger tested counts. All checks passed. The separate padding check uses $\varepsilon \in \{1/10000, 1/100, 1/10, 1/2, 99/100\}$ and confirms the formulas in Section 6; its five samples are distinct from the proof for the full interval $0 < \varepsilon < 1$.

Domination can be strict even when the global price exceeds one. The targeted diagnostic uses the independent weighted-variance formula for $\varphi(x) = x^2/2$. With one-based indices, take

$$x = (0, 1/36, 1/9, 1/4, 4/9, 25/36, 1), \quad p = (1, 1, 1, 2, 1/100, 1/10000, 1).$$

Common probability normalization changes no price. The cheapest pair is $\{5, 6\}$; a coarse flat optimum uses $\{4, 5\}$ and $\{6, 7\}$. Thus the selected quartet is $S = \{4, 5, 6, 7\}$, with the exact common flat costs

$$q = \frac{1}{323200}, \quad Q = \frac{1004419}{5210440992}.$$

Exact enumeration over all hierarchy shapes gives

$$R_7 = \frac{10314341331}{10144631900} \approx 1.016729 < \frac{2073082097481}{2018982631900} \approx 1.026795 = R_4(S).$$

The global optimum retains fine pair $\{5, 6\}$ and adds pair $\{1, 2\}$ at the coarse level. The quartet optimum instead merges $\{4, 5, 6\}$. This example was found after 32 constant-curvature candidates in the recorded search order. It directly rules out replacing the restriction inequality by equality for every individual source.

8 Interpretation, provenance, and remaining questions

The bound is uniform in source count because the last two capacities leave a deficit of only one and two cells. At general capacities a flat optimum can have many nonsingleton cells; the exchange classification used here no longer exhausts the possibilities. No claim is made about the sharp degree order for arbitrary prescribed capacities, or about one hierarchy that simultaneously approximates every capacity. For $N = 4$ only, adding the one-output and four-output levels is trivial; this fact does not extend the present theorem to general all-capacity hierarchies.

The centroid identity, scalar nearest-center contiguity, and Jensen kernels are classical Bregman facts [1]. The adjacent-triple inequality and complete polynomial quartet estimate are inherited from Phase 9 [3], while the endpoint lower construction originated in the earlier working paper [2]. Appendices A and B reproduce the needed arguments, so no lower kernel estimate or upper analytic dependency is left implicit. The new claims of this stage are the arbitrary- N quartet reduction, its envelope consequences, and the uniform exterior-anchor embedding. This description of project provenance is not a historical originality determination.

There is no minimum-mass or minimum-gap constraint in (1); a joint constrained envelope remains a different question. The bounded-loss normalization preserves ratios, and the explicit absolute-risk bounds tend to zero. Large multiplicative prices therefore give no nonvanishing absolute-risk separation. No neural training, sample-complexity, optimization-efficiency, or generalization claim is made. The internal review records and exact-arithmetic checks are supporting evidence, not external peer review or formal verification.

A Inherited weighted quartet analytic estimate

The following proposition, including its constants and proof strategy, is the finalized Phase 9 weighted quartet theorem. It is reproduced here so the upper-bound argument can be read without another file. It is not a new Phase 10 analytic estimate. The source is the weighted-quartet theorem in the Phase 9 working paper [3]; the exact source path and reviewed hash are recorded in the accompanying build receipt.

Proposition A.1 (Phase 9 weighted quartet estimate). *For four distinct weighted states and nonnegative nonzero polynomial curvature of degree at most $r \geq 1$, let R be the unrestricted deterministic hierarchy price at capacities $(2, 3)$. If $R \geq 128$, then*

$$R^2 \leq 16r^2 e^{20r/\sqrt{R}}.$$

If $R \geq 256$, then $R(\log R)^2 \leq 1936r^2$. In particular, for $r \geq 2$, $R \leq 1936r^2/(\log r)^2$.

Proof. We record two classical polynomial estimates. If $f \geq 0$ has degree at most r on an interval E of length L , and $M_E = \max_E f$, the classical Markov inequality [5] $\|f'\|_\infty \leq 2r^2 M_E/L$ implies

$$\int_E f \geq \frac{LM_E}{8r^2}. \quad (2)$$

Indeed a one-sided segment of length $L/(4r^2)$ is available from a maximizer, and $f \geq M_E/2$ there. If t_0 lies at distance $d > 0$ to the right of E , Chebyshev extremality gives

$$|f(t_0)| \leq M_E T_r(1 + 2d/L) \leq M_E e^{2r\sqrt{d/L}}. \quad (3)$$

For completeness, after rescaling E to $[-1, 1]$, any violation of the first inequality would yield $\lambda \in (0, 1)$ such that $T_r - \lambda f/M_E$ has r roots between the alternating extrema of T_r and an additional root at the external point. This contradicts its degree. The exponential bound follows from $T_r(\cosh s) = \cosh(rs)$ and $\operatorname{arcosh}(1+x) \leq \sqrt{2x}$.

Since $R > 1$, the argument of Theorem 3.1, specialized to four states, permits the notation

$$A < B < C < D, \quad q = D(BC) = \operatorname{OPT}_3, \quad Q = D(AB) + D(CD) = \operatorname{OPT}_2.$$

Write $p_L = D(AB)$, $p_R = D(CD)$, $T_L = D(ABC)$, and $T_R = D(BCD)$. Three feasible deterministic hierarchies give

$$R \leq \frac{\min(p_L, p_R)}{q}, \quad R \leq \frac{T_L}{Q}, \quad R \leq \frac{T_R}{Q}. \quad (4)$$

For the first retain $AB \mid CD$ and merge the cheaper outer pair at the fine level. For the others retain fine pair BC and merge the corresponding adjacent triple at the coarse level. All right sides are at least one by the defining flat optimality. No enumeration of noncontiguous optimal hierarchies is needed.

Reflect the source if needed so that the source weights satisfy $p_B \leq p_C$. Normalize only the coordinate of the triple BCD to

$$B = 0, \quad C = c \in (0, 1), \quad D = 1, \quad (p_B, p_C, p_D) = (u, v, w), \quad 0 < u \leq v.$$

The transformed curvature is $L_0^2 g(B + L_0 t)$, where L_0 is the original triple hull length; its degree and nonnegativity are preserved. Weights are unchanged. Denote that curvature again by g . The triple, adjacent-pair, and outer-pair kernels are

$$\begin{aligned} K(t) &= \min\{ut + v(t - c)_+, w(1 - t) + v(c - t)_+\}, \\ k_q(t) &= \min\{ut, v(c - t)\} \mathbf{1}_{[0, c]}(t), \\ k_p(t) &= \min\{v(t - c), w(1 - t)\} \mathbf{1}_{[c, 1]}(t), \\ H(t) &= \min\{ut, w(1 - t)\}. \end{aligned}$$

Put $T = T_R = \int K g$ and $S = p_R + q = \int (k_q + k_p) g$. From (4),

$$q \leq \frac{Q}{2R} \leq \frac{T}{2R^2}, \quad S \leq Q + q \leq \frac{2T}{R}. \quad (5)$$

Two pointwise kernel comparisons control degenerating weights:

$$0 \leq K - H \leq k_q + k_p, \quad (6)$$

and, in their respective open intervals,

$$\frac{k_q(t)}{K(t)} \geq \min\{1, (c - t)/t\} \quad (t < c), \quad \frac{k_p(t)}{K(t)} \geq \frac{t - c}{c + 2(t - c)} \quad (t > c). \quad (7)$$

For $t < c$, subtracting H from K amounts to adding $v(c - t)$ to one entry of a minimum, so the increment is between zero and $\min\{ut, v(c - t)\} = k_q$. For $t > c$, adding $v(t - c)$ to the other entry gives the bound by k_p . For the first ratio, use $K \leq ut$ and $v \geq u$. For the second, write $\Delta = t - c > 0$ and use

$$\frac{\min\{B, C\}}{\min\{A + B, C\}} \geq \frac{B}{A + B}, \quad A = ut, \quad B = v\Delta, \quad C = w(1 - t),$$

followed by $v \geq u$. Endpoint claims follow by continuity.

Assume $R \geq 128$ and set $\varepsilon = 16/R \leq 1/8$ and $J = \{t \in [0, 1] : |t - c| \leq \varepsilon c\}$. Outside J , (7) yields $k_q + k_p \geq \varepsilon K/(1 + 2\varepsilon)$. Consequently,

$$\int_{[0,1] \setminus J} Kg \leq \frac{1 + 2\varepsilon}{\varepsilon} S \leq \frac{5T}{32}, \quad \int_J Hg \geq T - \frac{5T}{32} - S \geq \frac{3T}{4}.$$

Here $S \leq 2T/R \leq T/64$. Since $H(t) \leq u(1 + \varepsilon)c$ on J and $|J| \leq 2\varepsilon c$, there is $t_0 \in J$ such that

$$g(t_0) \geq \frac{T}{4\varepsilon uc^2}.$$

Consider $E = [c/4, c(1 - 2\varepsilon)]$. It has length at least $c/2$; t_0 is to its right at distance at most $3\varepsilon c$; and $k_q \geq 2u\varepsilon c$ on E . Using (3),

$$M_E \geq \frac{T}{4\varepsilon uc^2} e^{-\sqrt{24}r\sqrt{\varepsilon}} \geq \frac{T}{4\varepsilon uc^2} e^{-20r/\sqrt{R}}.$$

Equation (2) now implies

$$q \geq 2u\varepsilon c \int_E g \geq \frac{u\varepsilon c^2 M_E}{8r^2} \geq \frac{T}{32r^2} e^{-20r/\sqrt{R}}.$$

Together with (5), this proves the implicit inequality.

Set $z = r/\sqrt{R}$. The inequality implies

$$\log(R/16) \leq 20z + 2 \log z \leq 22z.$$

For $R \geq 256$, the left side is at least $\frac{1}{2} \log R$, yielding $\sqrt{R} \log R \leq 44r$, hence $R(\log R)^2 \leq 1936r^2$. For $r \geq 2$ and $R \geq 256$ with $R > r$, use $\log R \geq \log r$. If $R \leq r$, use $(\log r)^2 \leq r$; if $R < 256$, use $(\log r)^2 \leq r^2$ and $256 < 1936$. These cases give the explicit bound. $\square$

B The matching endpoint lower construction

This appendix reproduces the Phase 8 endpoint construction [2], as already reproduced in Phase 9 [3], with its full kernel proof and bounded-loss normalization. It supplies the core used in Theorem 5.1. The source weights and posterior gaps are both allowed to vary with degree.

B.1 An endpoint-localized polynomial curvature

The only approximation ingredient needed is elementary external growth of Chebyshev polynomials. It is a classical polynomial localization mechanism; the identity $T_n(\cosh v) = \cosh(nv)$ is recorded in [4, Eq. 18.5.1]. Squared-Chebyshev endpoint needles and their exponential tail mechanisms also appear explicitly in Slot-Laurent [6, Section 4.1]; see also [7]. None of this polynomial technology is claimed as original here. The kernel estimates below are supplied in full.

For an integer n sufficiently large, put

$$d = d_n = \left(\frac{4 \log n}{n} \right)^2, \quad p_n(t) = T_n \left(\frac{2t - 1 - d}{1 - d} \right)^2, \quad Z_n = \int_0^1 p_n(t) dt, \quad \kappa_n(t) = \frac{p_n(t)}{Z_n}. \quad (8)$$

Here κ_n denotes its polynomial on the whole real line, although its normalization and probability-mass estimates refer only to $[0, 1]$. Its degree is $2n$ and it is nonnegative everywhere. For $t \in [d, 1]$ the argument of T_n belongs to $[-1, 1]$, so $p_n(t) \leq 1$. On $[0, d/2]$ the absolute value of that argument is at least $1 + d/(1 - d) \geq 1 + d$. For $0 \leq v \leq 1$, $\operatorname{arcosh}(1 + v) \geq \sqrt{v}$: for example, $\cosh(\sqrt{v}) \leq 1 + v$

follows by its power series. Using the parity of T_n and the hyperbolic identity therefore gives, for $d \leq 1/4$,

$$p_n(t) \geq \frac{1}{4}e^{2n\sqrt{d}} \quad (0 \leq t \leq d/2), \quad Z_n \geq \frac{d}{8}e^{2n\sqrt{d}}.$$

Consequently

$$\int_d^1 \kappa_n(t) dt \leq \tau_n, \quad \tau_n := 8d^{-1}e^{-2n\sqrt{d}} = \frac{1}{2n^6(\log n)^2} = o(d^3). \quad (9)$$

The normalized maximum also satisfies

$$\max_{[0,1]} \kappa_n \leq 32n^2. \quad (10)$$

For completeness, the rescaled classical Markov inequality [5] gives $\|\kappa'_n\|_\infty \leq 2(2n)^2M$ when $M = \|\kappa_n\|_\infty$. From a maximizer, one direction remains in $[0, 1]$ for length $1/(4(2n)^2)$, throughout which $\kappa_n \geq M/2$. Integrating and using $\int_0^1 \kappa_n = 1$ yields $M \leq 8(2n)^2$.

Define

$$g_n(t) = \kappa_n(t) + \kappa_n(1-t) + d^2, \quad \varphi''_n = g_n, \quad \varphi'_n(1/2) = \varphi_n(1/2) = 0. \quad (11)$$

This curvature is strictly positive on the whole real line and symmetric about $1/2$. Its degree is exactly $r = 2n$: both squared-polynomial terms have positive leading coefficient of the same even degree. The generator is a symmetric strictly convex polynomial of degree $2n + 2$.

B.2 All deterministic hierarchies are expensive

Take the sorted source locations and probabilities

$$(x_A, x_B, x_C, x_D) = (0, 2d, 1-2d, 1), \quad (p_A, p_B, p_C, p_D) = (a, b, b, a), \quad b = d^2, \quad a = \frac{1}{2} - b. \quad (12)$$

For all sufficiently large n , these are distinct and all weights are positive. Every distortion and flat denominator below uses these original weights, with no conditional renormalization of a cell.

For a pair $u < v$ of respective weights p_u, p_v , its exact Jensen curvature kernel is

$$K_{uv}(t) = \min\{p_u(t-u), p_v(v-t)\} \mathbf{1}_{[u,v]}(t). \quad (13)$$

For an arbitrary cell S , the kernel is

$$K_S(t) = \sum_{i \in S} p_i(x_i - t)_+ - p_S(m_S - t)_+, \quad D(S) = \int K_S(t) g_n(t) dt.$$

These formulas follow by writing a twice differentiable convex generator, up to an affine term, as an integral of $(x-t)_+$ against its curvature.

Set

$$P = D(AB) = D(CD), \quad q = D(BC), \quad T = D(ABC) = D(BCD), \quad J = D(AD).$$

The equalities use reflection symmetry. Since $128n^2b = 128n^2d^2 \rightarrow 0$, equations (9)–(10) imply, eventually,

$$I = [4b, d] \neq \emptyset, \quad \int_I \kappa_n(t) dt \geq 1 - \tau_n - 128n^2b \geq \frac{1}{2}. \quad (14)$$

We may also assume $d \leq 1/16$ and $\tau_n \leq d^2/2$.

The pair AB has mean $4bd$. Thus on I , which lies above that mean, $K_{AB}(t) = b(2d - t) \geq bd$. Conversely $K_{AB} \leq 2bd$ everywhere. On $[0, 2d]$, the reflected needle contributes at most τ_n to the integrated curvature, since $1 - 2d \geq d$. Hence

$$\int_0^{2d} g_n \leq 1 + \tau_n + 2d^3 \leq 2.$$

Together with (14), this proves

$$\frac{bd}{2} \leq P \leq 4bd. \quad (15)$$

For the middle pair, $K_{BC} \leq b/2$. The two needles contribute at most $2\tau_n$ on $[2d, 1 - 2d]$, and the constant curvature contributes at most d^2 . Therefore

$$0 < q \leq \frac{b}{2}(2\tau_n + d^2) \leq bd^2. \quad (16)$$

The triple ABC has mean $b/(1/2 + b) \leq 2b$. For $t \in I$ this mean lies below t , while both inner locations lie above t , so

$$K_{ABC}(t) = b(2d - t) + b(1 - 2d - t) = b(1 - 2t) \geq b/2.$$

The outer pair has kernel $K_{AD}(t) = a \min(t, 1 - t)$. On I this is at least $4ab \geq b$, because $a \geq 1/4$. Thus

$$T \geq b/4, \quad J \geq b/2. \quad (17)$$

We now cover all deterministic hierarchy choices, including noncontiguous ones. A hierarchy with fewer than the allowed numbers of cells can be refined to exactly three fine cells and two coarse cells without increasing either cost: split fine cells within their current coarse cells, then split the coarse level if necessary by grouping the three fine cells. Every exact three-cell partition of four states merges one pair. Besides BC , each pair has cost at least P . This is immediate for AB, CD ; the pair AC has the same endpoint weights as AB and a more distant right endpoint, so its kernel dominates K_{AB} pointwise, and BD is its reflection. Finally, $J \geq b/2 \geq P$ by (15) and $d \leq 1/16$.

If the fine pair is not BC , then $\text{OPT}_3 \leq q$ gives fine-level ratio at least $P/q \geq 1/(2d)$. If the fine pair is BC , its exact two-cell coarsenings are $ABC \mid D$, $A \mid BCD$, and $AD \mid BC$. By (17) each costs at least $b/4$. Since the available coarse split $AB \mid CD$ gives $\text{OPT}_2 \leq 2P \leq 8bd$, this hierarchy has coarse-level ratio at least $1/(32d)$. These two cases prove

$$\rho_{\varphi_n, \text{det}} \geq \frac{1}{32d_n}, \quad \rho_{\varphi_n, \text{stoch}} \geq \frac{1}{64d_n}, \quad (18)$$

the second inequality following from Lemma 2.3. Notice that this proof needs neither exact flat optimizer identification nor an assertion that contiguous deterministic hierarchies exhaust the optimum.

Theorem B.1 (Matching arbitrary-weight four-state lower bound). *Let $\hat{C}_r^{\text{det}}$ and $\hat{C}_r^{\text{stoch}}$ denote the two/three-output four-state degree envelopes in which all positive source weights are allowed. There is an absolute $c > 0$ such that, for all sufficiently large r ,*

$$\hat{C}_r^{\text{det}} \geq \hat{C}_r^{\text{stoch}} \geq c \frac{r^2}{(\log r)^2}.$$

Consequently no absolute-constant $O(r/\log r)$ bound holds in this weighted four-state setting. The witnesses can have all posteriors in $[1/4, 3/4]$ and polynomial generators inducing smooth strictly proper losses bounded in $[0, 1/2]$.

Proof. Equations (8) and (18) give the claim at degrees $r = 2n$, since $1/d_n = n^2/(16(\log n)^2)$. For a degree-at-most- r envelope take $n = \lfloor r/2 \rfloor$.

To put all posteriors in the stated interior interval, use $y = 1/4 + x/2$. The four locations become $(1/4, 1/4 + d, 3/4 - d, 3/4)$. Let $M_n = \max_{t \in [-1/2, 3/2]} g_n(t)$ and set

$$\tilde{\varphi}_n(y) = \frac{\varphi_n(2y - 1/2)}{4M_n}.$$

The fact that g_n is a sum of squares plus d^2 on the *whole real line* ensures $0 < \tilde{\varphi}_n'' \leq 1$ on $[0, 1]$. This affine coordinate change and positive rescaling preserve every price and curvature degree. The generator remains symmetric about $1/2$. Put $h(y) = \tilde{\varphi}_n(0) - \tilde{\varphi}_n(y)$ and define

$$\ell(y, 1) = h(y) + (1 - y)h'(y), \quad \ell(y, 0) = h(y) - yh'(y).$$

Expected regret is $B_{\tilde{\varphi}_n}$, so the loss is strictly proper. Symmetry and the supporting-line inequalities make both losses nonnegative; their derivatives are $-(1 - y)\tilde{\varphi}_n''(y)$ and $y\tilde{\varphi}_n''(y)$, respectively. Their common maximum is $\tilde{\varphi}_n'(1) = -\tilde{\varphi}_n'(0) = \int_{1/2}^1 \tilde{\varphi}_n'' \leq 1/2$. $\square$

The smallest mass in this family is $d_n^2 = 256(\log n)^4/n^4$. The generators, weights, and posterior gaps all vary with n . This does not determine a sharp joint minimum-mass/degree envelope. The normalization preserves price, not absolute excess cost; explicit vanishing flat-cost bounds for the embedded family appear in Section 5.

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